User Guide: Enterprise AI Decision Intelligence Platform

This guide explains how business users and technical users should use the application.

1. What The App Does

The application helps teams review AI/ML decision models, understand their business impact, and inspect the technical evidence behind each run.

It is organized around four model workflows:

- TensorFlow Next Best Action Model
- TensorFlow Churn Propensity Model
- Bayesian Media Mix Optimization
- Causal Incrementality Model

Each workflow moves from model inputs to recommendation, impact forecast, decision evidence, activation output, and governance evidence.

2. Recommended Presentation Flow

Use this sequence for an end-to-end walkthrough:

- Start at AI Decision Portfolio.
- Open the Model Decision Workbench.
- Walk through model inputs, recommendation, impact forecast, and decision evidence.
- Open AI Impact Summary to show portfolio-level health.
- Open AI/ML Platform Architecture to explain the technical runtime and evidence path.

3. Starting The App

From the repository root:

```
make standalone
```

Open:

```
http://127.0.0.1:8080/ui/experience/index.html
```

If the environment is already prepared:

```
make standalone-fast
```

For live UI/API mode:

```
make ui-live
```

4. Main Pages

Page	URL	Primary Audience
AI Decision Portfolio	/ui/experience/business-home.html	Business and executive users
AI Impact Summary	/ui/experience/business-exec-summary.html	Executives and business owners
Model Decision Workbench	/ui/experience/workbench.html	Business users and technical reviewers
AI/ML Platform Architecture	/ui/experience/index.html	Architects, ML engineers, and platform teams

5. Business User Guide

Business users should focus on the decision story and model impact.

Step 1: Open AI Decision Portfolio

Use the portfolio page to understand which models are available and what each model is designed to decide.

Look for:

- Model name
- Decision output
- Primary business metric
- Recommended presentation sequence

Step 2: Open A Model Workflow

Click Open Model Workflow for one of the models.

Recommended first workflow:

```
TensorFlow Next Best Action Model
```

Step 3: Walk The Business Steps

In the Model Decision Workbench, use the left-side step navigation:

- Model Inputs
- Model Recommendation
- Impact Forecast
- Decision Evidence

Use these questions while presenting:

- What customer, campaign, or spend signals entered the model?
- What does the model recommend?
- What impact does the model forecast?
- What evidence supports the decision?

Step 4: Review The AI Impact Summary

Open AI Impact Summary to see portfolio-level model health.

Look for:

- Models reviewed
- Models healthy
- Average pipeline pass rate
- Latest run status
- Model-specific business KPIs

Step 5: Share Or Print

Use Print / Save PDF on the AI Impact Summary page when you need an executive-ready output.

6. Technical User Guide

Technical users should focus on architecture, run evidence, artifacts, and governance.

Step 1: Open AI/ML Platform Architecture

Use the architecture page to explain the end-to-end platform.

The core architecture is:

```
Data Sources
-> Ingestion + Event Bus
-> Raw Storage + Curated Warehouse
-> Identity Resolution + Customer 360
-> Feature Layer
-> Model Layer
-> Serving + Activation
-> Monitoring + Governance
```

Step 2: Walk The Eight ML Stages

Use the side navigation on the architecture page to walk stage by stage.

For each stage, review:

- What the stage does
- Data in
- Data out
- Components
- Evidence artifacts

Step 3: Use Model Evidence View

Open Model Decision Workbench and select Model Evidence.

Use this view to inspect:

- Contract controls
- Stage inputs and outputs
- Model backend
- Model metrics
- Activation payloads
- Validation gates
- Artifact paths

Step 4: Inspect Model Evidence Center

On the architecture page, open Model Evidence Center.

Review:

- Model Evidence
- AI/ML Architecture Flow
- Model Run History
- AI Portfolio Health

Step 5: Run A New Model Workflow

Use Model Execution controls on the architecture page.

Common options:

- Select an AI/ML use case.
- Choose Standalone AI Runtime or Integrated AI Runtime.
- Choose runtime mode.
- Select a scenario preset if needed.
- Click Run Standalone AI Runtime or Run Integrated AI Runtime.

Step 6: Simulate Stage Execution

Use the simulation controls to explain the pipeline gradually:

- Run Next Stage
- Pause Simulation
- Resume Simulation
- Reset Simulation

This is useful when presenting the architecture to technical audiences.

7. Runtime Profiles

```
Profile | Use When
Standalone AI Runtime | You want the app to run locally without external services.
Integrated AI Runtime | You want to show optional Kafka, Postgres, or MinIO-style
infrastructure mirroring.
Enterprise Integration Profile | You want to show readiness for MLflow, Feast, Splink,
Airflow, Keycloak, monitoring, and metrics.
```

The app can run standalone without deployment dependencies. Enterprise integrations are optional.

8. Common Tasks

Run all model workflows

```
python3 -m pipelines.cli run-stack-all
```

Run one workflow

```
python3 -m pipelines.cli run-stack --use-case UC-NBA-RET-001
```

Run synthetic-only mode

```
python3 -m pipelines.cli run-stack --use-case UC-NBA-RET-001 --runtime-mode
synthetic_only
```

View recent runs

```
python3 -m pipelines.cli list-runs
```

Inspect run stages

```
python3 -m pipelines.cli show-stages --use-case UC-NBA-RET-001 --run-id <run_id>
```

Inspect model readiness

```
python3 -m pipelines.cli model-readiness --use-case UC-NBA-RET-001 --run-id <run_id>
```

9. Troubleshooting

If the app is not loading:

- Confirm the server is running.

- Confirm the URL uses the right port.
- Run make standalone-fast if dependencies and artifacts already exist.
- Run make smoke-standalone to validate the app.

If model evidence is missing:

- Run at least one workflow.
- Refresh the page.
- Check artifacts/<use_case_id>/<run_id>/summary.json.

If integrated runtime is unavailable:

- Use Standalone AI Runtime.
- Confirm optional infrastructure is running before using Integrated AI Runtime.

10. What To Show In A Meeting

For business stakeholders:

- AI Decision Portfolio
- One Model Workflow
- AI Impact Summary

For technical stakeholders:

- AI/ML Platform Architecture
- Eight ML stages
- Model Evidence View
- Model Run History
- Runtime Evidence

For platform leaders:

- Business value from model workflows
- Evidence-backed architecture

- Standalone runtime
- Enterprise integration path